1.

HW2_1.cpp

```cpp
#include<iostream>
using namespace std;

float purchase,cash;
int change,dollar,quarter,dime,nickel,penny;

int main()
{
    cout << "Enter the purchase amount:";
    cin >> purchase;
    cout << "Enter the cash amount:";
    cin >> cash;
    change = (cash-purchase)*100;


    dollar = change/100;
    change = change%100;

    quarter = change/25;
    change = change%25;

    dime = change/10;
    change = change%10;

    nickel = change/5;
    change = change%5;

    penny = change/1;
    change = change%1;


    cout << "Change:" << endl;
    cout << dollar << " dollars\n" << quarter << " quarters\n" << dime << " dimes\n" << nickel << " nickels\n" << penny << " pennies";

    return 0;
}
```

```
Enter the purchase amount:120.47
Enter the cash amount:150
Change:
29 dollars
2 quarters
0 dimes
0 nickels
3 pennies
--------------------------------
Process exited after 13.59 seconds with return value 0
請按任意鍵繼續 . . .
```

2.

HW2_2.cpp

```cpp
#include<iostream>
using namespace std;

float celsius,fahrenheit;

main()
{
    cout << "Enter the temperature in Celsius:";
    cin >> celsius;
    fahrenheit = celsius*9/5+32;
    cout << celsius << " Celsius\n" << fahrenheit << " Fahrenheit";
    return 0;
}
```

```
Enter the temperature in Celsius:36
36 Celsius
96.8 Fahrenheit
--------------------------------
Process exited after 4.555 seconds with return value 0
請按任意鍵繼續 . . .
```

3.

HW2_3.cpp

```cpp
#include<iostream>
#include<iomanip>
using namespace std;

float checkValue,shareCharge1,shareCharge2,shareCharge3;

main()
{
    cout << "Enter check value:";
    cin >> checkValue;
    checkValue *= 1.295;
    shareCharge1 = checkValue/36*14;
    shareCharge2 = checkValue/36*11;
    shareCharge3 = checkValue/36*11;

    cout << "Total check after tax and tip :" << checkValue << endl;
    cout << "Family 1 pays:" << fixed << setprecision(2) << shareCharge1 << endl;
    cout << "Family 2 pays:" << fixed << setprecision(2) << shareCharge2 << endl;
    cout << "Family 3 pays:" << fixed << setprecision(2) << shareCharge3;

    return 0;
}

```

```
Enter check value:150
Total check after tax and tip :194.25
Family 1 pays:75.54
Family 2 pays:59.35
Family 3 pays:59.35
--------------------------------
Process exited after 2.89 seconds with return value 0
請按任意鍵繼續 . . .
```

4.

HW2_4.cpp

```cpp
#include<iostream>
#include <iomanip>
using namespace std;

int tv,dvd,control;
float tvSets,dvdPlayers,remoteControllers,subtotal,tax,total;

main()
{
    cout << "How many TVs were sold?";
    cin >> tv;
    tvSets = tv*1400;
    cout << "How many DVDs were sold?";
    cin >> dvd;
    dvdPlayers = dvd*220;
    cout << "How many remote controllers were sold?";
    cin >> control;
    remoteControllers = control*35.2;

    subtotal =  tvSets + dvdPlayers + remoteControllers;
    tax = subtotal*0.0825;
    total = subtotal+tax;

    cout << "Quantity\tDescription\tUnit Price\tTotal Price" << endl;
    cout << tv << "\t\tTV" << "\t\t1400\t\t" << tvSets << endl;
    cout << dvd << "\t\tDVD" << "\t\t220\t\t" << dvdPlayers << endl;
    cout << control <<"\t\tRemote" << "\t\t35.2\t\t" << remoteControllers << endl;

    cout << "Total before tax\t\t\t\t" << subtotal << endl;
    cout << "Tax\t\t\t\t\t" << fixed << setprecision(2) << tax << endl;
    cout << "Total\t\t\t\t\t" << total;
    return 0;
}
```

```
How many TVs were sold?2
How many DVDs were sold?3
How many remote controllers were sold?4
Quantity        Description     Unit Price      Total Price
2               TV              1400            2800
3               DVD             220             660
4               Remote          35.2            140.8
Total before tax                                3600.8
Tax                                             297.07
Total                                           3897.87
--------------------------------
Process exited after 3.778 seconds with return value 0
請按任意鍵繼續 . . .
```